MCQ Test for 41107 Marine and Ocean engineering (2025)

[THIS IS A TEST EXAMINATION SET]

The scoring algorithm is based on "One best answer"

This means:

There is always only one correct answer
Students are only able to select one answer per question
Every correct answer corresponds to 1 point
Every incorrect answer corresponds to 0 points (incorrect answers do not result in subtraction of points)

<u>If not specifically stated you may assume the following:</u>
Water density, $\rho = 1025 \ \mathrm{kg/m^3}$
Kinematic viscosity of water, $\nu = 1 \cdot 10^{-6} \ \mathrm{m^2/s}$
Cylinder diameter, D
Steady current velocity, U_c
Maximum horizontal velocity of oscillatory flow, U_m
Deep-water conditions.
Long-crested waves.
Processes of waves and wave-induced responses are Gaussian with the values of the peaks following from the Rayleigh distribution.

SHIP MOTION DYNAMICS: Transfer functions of a conventional container ship are available. At a given frequency $\omega = 0.6$ rad/s, the real and imaginary parts of the pitch transfer function is (Re,Im) = (-5.0e-2, 8.0e-2) rad/m for wave heading 150°. What is the phase lag ϕ (relative to the wave) of pitch at the frequency $\omega = 0.6$ rad/s?

Choose one answer

- $\phi = 238°$
- $\phi = 58°$
- $\phi = 122°$

WAVES: A ship sails with a speed of 18 knots. The ship encounters regular waves at an angle beta = 135° (bow-quartering waves). *On* the ship, the encountered wave period is T_{enc} = 10 s. What is the absolute (true) wave period T, as measured by a fixed observer relative to the waves?

Choose one answer

- T = 12.5 s
- T = 13.0 s
- T = 13.5 s

WAVES: From a wave record consisting of 141 waves (trough to crest), the number of waves of various heights are as reported in the table below. What is the significant wave height?

Height [m]	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0
No. of waves	11	38	45	27	12	4	3	1

Choose one answer

- ○ $H_S = 2.25$ m
- ○ $H_S = 2.15$ m
- ○ $H_S = 2.35$ m

SHIP MOTION DYNAMICS: A vertical circular cylinder floats with its longitudinal axis vertically aligned. The cylinder undergoes free oscillations (pure heave); damping is neglected. The cylinder has diameter D = 0.3 m and floats at a draught h = 1.1 m measured from the bottom of the cylinder, when at rest in still water. At which frequency ω_N does the cylinder oscillate?

Choose one answer

- ω_N > 3 rad/s
- ω_N = 3 rad/s
- ω_N < 3 rad/s

VORTEX INDUCED VIBRATIONS. Plane oscillatory flow around a free, smooth, flexibly-mounted, circular cylinder. What is the reduced velocity when the oscillatory flow is defined by $U[\mathrm{m/s}] = 2.5 \cdot \sin(\omega t)$, $D = 0.5\ \mathrm{m}$, and the angular natural frequency $\omega_n = 7.85\ \mathrm{rad/s}$?

Choose one answer

- $V_r = 6.25$
- $V_r = 0.25$
- $V_r = 4.0$

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free, smooth circular cylinder. Consider the case defined by $U_m = 1 \text{ m/s}, D = 1 \text{ m}$ and the wave period $T = 5 \text{ s}$. Further, assume the following functional expression for determining C_D and C_M. $C_D = 1.3 + 0.1(KC - 12)$ and $C_M = \min(2.0; 2.0 - 0.044(KC - 3))$ for $2 < KC < 12$. What is the ratio between the maximum inertia and maximum drag force ($F_{I,max}/F_{D,max}$)?

Choose one answer

- 0.16
- 4.00
- 6.29

SPECTRAL ANALYSIS: A wave system is described by a Bretschneider wave spectrum with H_s = 2 m and T_p = 10 s. A small boat of length L = 8 m is sailing in the wave system. What is the significant amplitude h_{heave} of the boat's heave motion approximately?

Choose one answer

- h_{heave} = 4 m
- h_{heave} = 1 m
- h_{heave} = 2 m

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free cylinder: What is the pressure in excess of hydrostatic pressure in the stagnation point?

Choose one answer

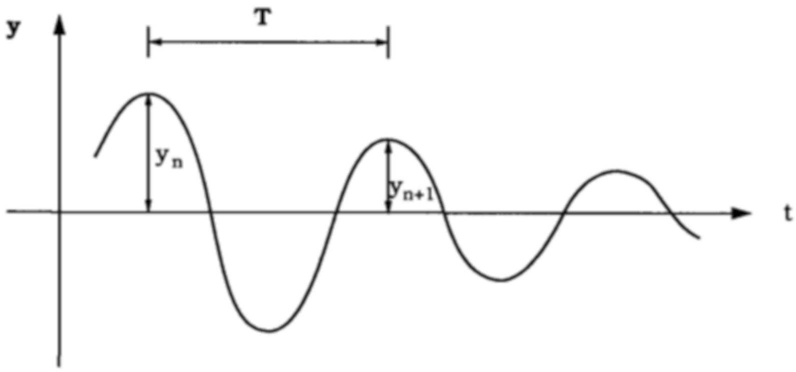
- $c_p = 0$
- $\frac{1}{2}\rho U_c^2$
- 1 Pa

VORTEX INDUCED VIBRATIONS. Calculate the un-damped natural frequency of a flexibly mounted circular cylinder in water where the following is given: the cylinder diameter $D = 0.2\ \mathrm{m}$, the mass (per unit length) $m = 4\ \mathrm{kg/m}$, the spring constant (per unit length) $k = 900\ \mathrm{kg/m}$ and it is known that the normalized cross-flow vibration amplitude $A/D < 0.8$.

Choose one answer

- $f_n = 0.58\ \mathrm{Hz}$
- $f_n = 2.39\ \mathrm{Hz}$
- $f_n = 0.79\ \mathrm{Hz}$

VORTEX INDUCED VIBRATIONS. The amplitude response as function of time in a free decay test in water is shown in the following figure. What is the total damping factor (ζ) and the un-damped natural angular frequency (ω_n) when $y_n = 2$, $y_{n+1} = 0.5$ and $T = 1$ s?



Choose one answer

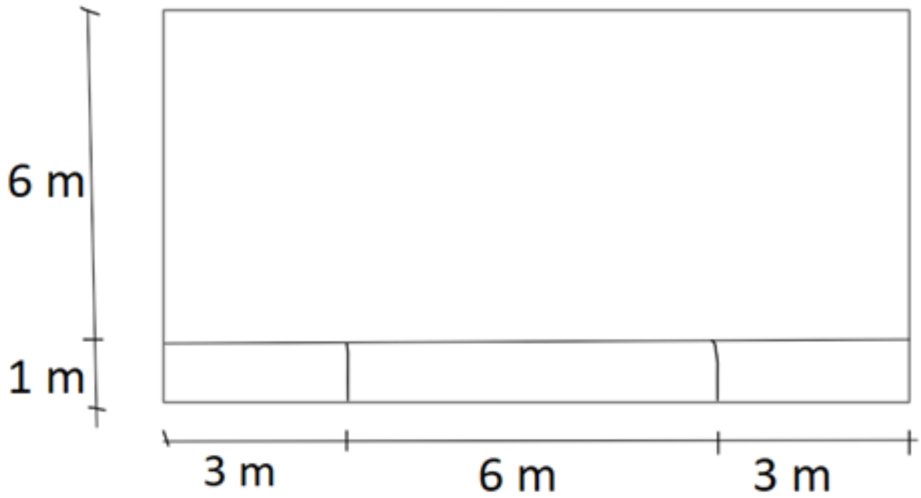
- $\zeta = 0.096$; $\omega_n = 6.25$ rad/s
- $\zeta = 0.318$; $\omega_n = 5.96$ rad/s
- $\zeta = 0.221$; $\omega_n = 6.13$ rad/s

SHIP HYDROSTATICS: A homogeneous log of square cross section, side $b = 1$ m, and length $l = 6$ m long floats with its length parallel to the water surface and with its cross sectional sides horizontally and vertically aligned as illustrated below; note that the indicated draught is *not* correct. The log's density is half that of the water. What happens to the log if it is given a small heeling moment which is instantly applied and released? Stillwater conditions are assumed.

Choose one answer

- ◯ The log is stable in its initial equilibrium position and will return to this position after the heeling moment is applied and released.
- ◯ The log is in an unstable equilibrium in its initial equilibrium position and will find a new equilibrium position after the heeling moment is applied and released.
- ◯ The log remains in the heeled position that it reaches after the heeling moment is applied and released since the log is unconditionally stable.

STRENGTH OF SHIPS: The cross section of a barge is shown below. All plate thicknesses are $t = 10$ mm. What is the vertical distance z_n from the keel to the neutral axis?



Choose one answer

- ○ $z_n = 2.72$ m
- ○ $z_n = 2.81$ m
- ○ $z_n = 2.95$ m

SPECTRAL ANALYSIS: Information used for spectral calculations of a ship is available. Specifically, it is known that, at the frequency ω_a, the wave energy spectral density is S(ω_a) = 2.3 m^2s/rad and the corresponding pitch energy spectral density is R(ω_a) = 0.006 rad^2s/rad. What value, approximately, has the magnitude $|\Phi(\omega_a)|$ of the pitch transfer function at the frequency ω_a?

Choose one answer

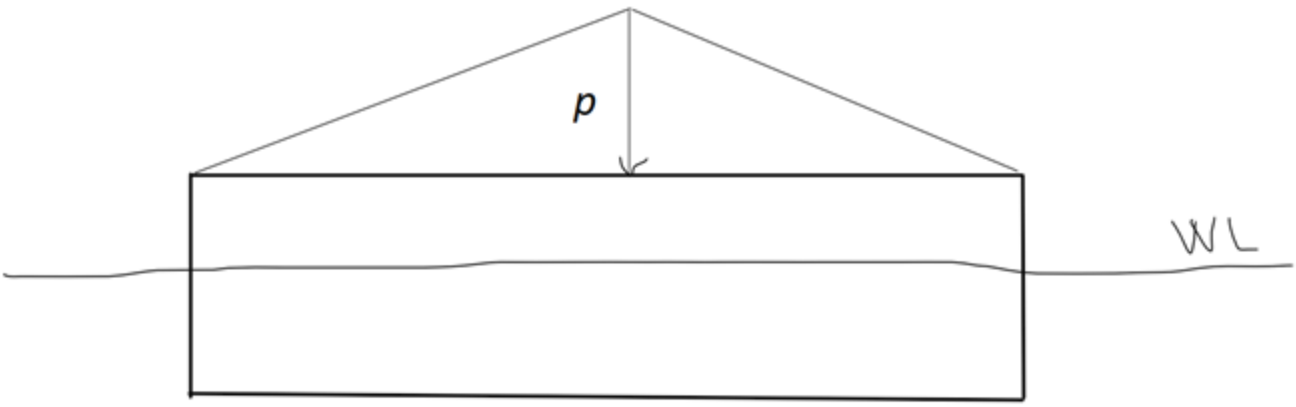
- $|\Phi(\omega_a)|$ = 2.9 deg/m
- $|\Phi(\omega_a)|$ = 1.3 deg/m
- $|\Phi(\omega_a)|$ = 0.15 deg/m

VORTEX INDUCED VIBRATIONS. Steady current around a free, flexibly-mounted circular cylinder. What is the ratio between the maximum in-line vibration amplitude and the maximum cross-flow vibration amplitude, A_x/A_y ? (values below 1 indicate cross-flow vibration amplitude is larger than in-line vibration amplitude)?

Choose one answer

- Approximately 1/10
- Approximately 10
- Approximately 1

STRENGTH OF SHIPS: The diagram below shows the distributed weight on a rectangular, box-shaped barge loaded with gravel. The barge is made from steel and its own weight is uniformly distributed along the length with intensity 0.1*p*. The barge's length and breadth are 50 m and 8 m, respectively, and it floats at the draught 4 m in freshwater (no heel and no trim). Approximately, what value has *p*?



Choose one answer

- p = 53 tons
- p = 38 tons
- p = 45 tons/m

SHIP RESISTANCE: A ship of mass 5,000 tons, length 120 m, is to be represented by a model 3 m long for resistance estimation. If performance is compared at the corresponding speeds, what will the ratio r of ship to model resistance approximately be?

Choose one answer

- $r = 6e04$
- $r = 6e03$
- $r = 6e05$

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free circular cylinder: Consider the following three cases defined by D and U_c. You can assume $C_D = 1.2$ in the sub-critical flow regime, 0.6 in the super-critical flow regime and 0.9 in the trans-critical flow regime. Which case has the highest drag force?

Choose one answer

- $D = 0.2 \text{ m}, U_c = 0.5 \text{ m/s}$
- $D = 5.0 \text{ m}, U_c = 0.2 \text{ m/s}$
- $D = 0.5 \text{ m}, U_c = 0.2 \text{ m/s}$

SHIP HYDROSTATICS: A rectangular, box-shaped barge of length L = 50 m and beam B = 7 m floats at a uniform draught T. What must be the draught for the transverse metacentre (M_T) to lie in the waterplane?

Choose one answer

- T = 3.1 m
- T = 2.9 m
- T = 2.7 m

FLOW AROUND AND FORCES ON CYLINDERS. A circular cylinder is placed on the seabed. $D = 0.5\ \mathrm{m}$, water depth $h = 12\ \mathrm{m}$, wave height $H = 1\ \mathrm{m}$ and wave period $T = 7\ \mathrm{s}$. What is the KC number?

Choose one answer

- $KC = 6.3$
- $KC = 4.2$
- $KC = 5.5$

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free smooth circular cylinder. The normalized fundamental lift frequency is $N_L = 3$. Which name is given to this flow regime?

Choose one answer

- three pairs, vortex shedding regime
- double pair, vortex shedding regime
- single pair, vortex shedding regime

SHIP RESISTANCE: In resistance estimation via model tests, it is important to ensure what?

Choose one answer

- ◯ Identical Reynolds numbers as well as Froude numbers of the full-scale and model-scale ships.
- ◯ Identical Froude numbers of the full-scale and model-scale ships.
- ◯ Identical Reynolds numbers of the full-scale and model-scale ships.

FLOW AROUND AND FORCES ON CYLINDERS. The wave period is T. What is the definition of the KC number?

Choose one answer

- ○ $KC = U_m D/\nu$
- ○ $KC = 0.44/(D/L)$
- ○ $KC = U_m T/D$

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free smooth circular cylinder, the wake is turbulent and the vortex shedding frequency, $f_v = 0.018\ \mathrm{Hz}, U_c = 0.2\ \mathrm{m/s}$, and $D = 5\ \mathrm{m}$. Which flow regime describes the flow?

Choose one answer

- Trans-critical
- Super-critical
- Sub-critical

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free circular cylinder. f_v is the vortex shedding frequency. What is the frequency of the oscillating drag force?

Choose one answer

- $1/2 f_v$
- f_v
- $2 f_v$